

Examiner reports

Q1.

Many candidates showed that they are competent with implicit differentiation and scored full marks here or at least some of the marks. The most common error was for candidates not to differentiate the 4 on the right-hand side, leaving it as 4, having got the left-hand side correct. The derivative of $2y^2$ was usually correct, but errors such as an extra x or y or a coefficient error crept into the derivative of $3xy$.

Q2.

Some candidates did not attempt part (a). However, those who could square $9 - 3x$ usually earned the marks. A surprising number substituted for both x and y and most were unable to complete the solution successfully.

In part (b)(i), the differentiation was carried out very well by most candidates. Errors in the value of k occurred due to taking out a common factor either before or after differentiating, so values of k such as 3 and 9 were common. A few weak candidates did not recognise that they had to differentiate V here, and generally were unable to proceed through the rest of the question.

In part (b)(ii), the factorisation was usually carried out successfully, and candidates who had not found the correct value of k were still able to earn these marks.

The most common error in part (c) was differentiating $x^2 - 4x + 3$, rather than $27(x^2 - 4x + 3)$, and this approach earned a method mark only.

Credit was given in part (d)(i) for substituting their values of x into their second derivative and so most candidates earned this mark, provided they had two values of x .

In part (d)(ii), most realised that $x = 1$ needed to be selected since $\frac{d^2y}{dx^2}$ was negative when $x = 1$.

In part (d)(iii), not all substituted $x = 1$ into the equation for V . Those who did often made arithmetic errors. Some gave the value of y when $x = 1$ and others gave the answer as

-54 , the value of $\frac{d^2y}{dx^2}$ when $x = 1$.

Q3.

Although many candidates were able to correctly write $\sqrt{x^3}$ in the form $x^{\frac{1}{2}}$ it was not uncommon to see the incorrect answers $k = 3.5$ or $k = \frac{1}{3}$. Although follow through marks were available, a significant minority of candidates subsequently differentiated $x^{\frac{1}{3}}$

incorrectly as $\frac{1}{3}x^{\frac{2}{3}}$. In addition to arithmetical errors the most common error in finding

the equation of the tangent was to use the gradient of the tangent as $-\frac{1}{5}$; in effect, the equation of the normal was found. Very few candidates failed to write their equation in the required form.

Q4.

Candidates generally applied the trapezium rule correctly and gave the final answer to the given degree of accuracy. Many candidates failed to gain full marks in part (b)(ii) as their diagrams did not show four trapeziums. There were also a significant number of candidates who either failed to draw any diagram or who drew the sloping sides of their trapeziums under the curve. Most candidates realised that the geometrical transformation was a stretch, but the wrong answer 'stretch parallel to the y -axis, scale factor 3' appeared almost as frequently as the correct answer. A high proportion of candidates showed that they were able to use logarithms to solve the equation $6^{3x} = 84$ but some only gave their answer to two decimal places. The final part of the question was poorly answered, although a minority of candidates did score some credit for partially correct answers, for example, $f(x) = 6^{x+1} - 2$.

Q5.

Part (a) of this question was reasonably answered with most candidates obtaining the method mark for the product rule. Some candidates then lost the accuracy mark through incorrect evaluation of the constant associated with the derivative of $\cos 2x$; 2 and $\frac{1}{2}$ were frequently seen.

Answers to part (b)(i), although frequently correct, were often badly set out with the function being equated to zero or x being changed to α at various points in the solution. Division of $\cos 2x$ by $\cos 2x$ often resulted in zero before going on to fudge the correct answer. Part (b)(ii) was usually well answered with correct evaluations of $f(0.4)$ and $f(0.5)$. Some candidates then lost marks by just saying α was 'between these two values' without stipulating 0.4 and 0.5. Part (b)(iii) was usually well done but there were many cases when

the tan became removed from its $2x$ and $\frac{1}{2}x \tan^{-1} = x$ was often seen. Part (b)(iv) was generally well answered but marks were lost from the use of degrees and for not writing the answer to the required degree of accuracy.

Many fully correct answers and many answers which only lost the final accuracy mark were seen in part (c). Other answers which had the wrong coefficient associated with the $\sin 2x$ often got the method marks. There were candidates who started with $u = x$ and $v =$

$\cos 2x$ or started with $\frac{dv}{dx} = x$ and obtained a more difficult integral.

Q6.

The vast majority of candidates showed they understood the requirements of this question although often algebraic errors were made.

Many candidates could not find the derivative of $\frac{1}{2t}$ correctly in part (a), the 2 moving between the numerator and denominator position to various powers; the derivative of $\frac{1}{t}$ itself was usually correct. However, most candidates used the chain rule correctly to find a gradient. Some candidates actually found a cartesian equation before differentiating, with some completing successfully.

In part (b) most candidates went on to use their gradient to find that of the normal, and bar a few slips in calculating the x and y coordinates, were successful in finding a correct equation, using their gradient.

In part (c) most candidates found t from $x = 4t + 3$ and substituted in $y = \frac{1}{2t} - 1$, often without any simplification, which was quite acceptable. Those who did attempt some simplification often made algebraic errors, but these were condoned following a correct substitution. Those who attempted an alternative method of eliminating t were more prone

to error. For some candidates, poor presentation led to $\frac{1}{2t}$ becoming $\frac{1}{2}t$.

Q7.

Part (a) Candidates did not seem confident at working on this kind of problem and algebraic weaknesses were evident. Many worked backwards from the result in part (a)(i) and did not always convince the examiner that they were considering the surface area of four faces and the base. The inability of most candidates to rearrange the formula to make h the subject in part (a)(ii) was alarming. Consequently few, without considerable fudging, could establish the printed formula for the volume. Part (b) Basic differentiation is well

understood and most candidates found $\frac{dV}{dx}$ correctly. Some tried to substitute $x = 3$ into the expression for V in order to show there was a stationary point, but usually this part was answered well.

Part (c) It was not uncommon to see the second derivative as $4x$ even though the first derivative was correct. A generous follow through was given here provided candidates could interpret the value of their second derivative.

Q8.

Part (a)(i) Some candidates ignored the request to state the coordinates of B even though they were using the height of the triangle as 5. The negative x -coordinate of A caused quite a few to feel that the triangle had a negative area. Far too many when finding

$\frac{1}{2} \times 1 \times 5$
wrote the answer as 3.

Part (a)(ii) Practically every candidate found the correct integral although some made errors when cancelling fractions.

Part (a)(iii) It was necessary here to have the lower limit as -1 and the upper limit as 0 .

Many reversed the order and by some trickery arrived at a positive value. This was penalised and so very few, even though many had an answer of 1 for the area, scored full marks for this part of the question.

Part (b) Most candidates differentiated correctly but, because of poor understanding of negative signs, many wrong values of -13 were seen for the gradient. There is obviously confusion for many between tangents and normals and several thought the gradient of the

tangent was $\frac{-1}{17}$.

Q9.

This calculus question again proved to be a good source of marks for many candidates.

Part (a)(i) was answered very well but in part (a)(ii) a significant minority used $\frac{d^2x}{dx^2} = 0$

instead of $\frac{dy}{dx} = 0$ dx^2 . In part (a)(iii), although most candidates showed that they understood the method to find the gradient of the tangent, a significant proportion failed to find and use the gradient of the normal. Many candidates showed that they had a good understanding of integration but, surprisingly, errors in integrating $(x + 1)$ were almost as

common as errors in integrating $\frac{4}{x^2}$. Examiners expected to see some simplification of

the algebraic term $\frac{+4x^{-1}}{-1}$. In the final part, the evaluation of $F(4) - F(1)$ as 8.5 was the most common error.

Q10.

In part (a)(i), many candidates were unable to produce a correctly expressed solution, despite dealing correctly with the differentiation. The main error was the omission of brackets around $8x + 3$. Part (a)(ii) was well answered by the majority of candidates. Part (b)(i) was well

answered by the majority of candidates, although a common error was $6y^2 + \frac{1}{x}$. Most candidates correctly substituted $y = 1$ and obtained 7, although 6 was a common evaluation; a substitution of $y = 2$ was also seen. Many candidates who obtained 7 then proceeded, incorrectly, to write $(y - 1) = 7(x - 2)$ although a few candidates correctly wrote

$\frac{dx}{dy}$ “ $(x - 2) = 7(y - 1)$ ”. The significance of $\frac{dx}{dy}$ was rarely appreciated, with a significant

number of candidates finding an incorrect gradient of $-\frac{1}{7}$, and confusing $\frac{dx}{dy}$ with the gradient of a normal.

Q11.

Part (a)(i) was well answered, with the main error being $\int -e^x dx = -2e^{2x}$. In part (a)(ii) the answer was given. Frequently candidates tried to 'fudge' their solutions to arrive at the correct result, with common error being $4 \ln 2 - 2 - \frac{1}{2} = 4 \ln 2 - \frac{3}{2}$, the given answer.

Part (b)(i) was again well answered by the majority of candidate, although in part (b)(ii) the step from $2x = \ln 4$ to obtain $x = \ln 2$ was often unconvincing. Some candidates answered part (c) very well and gained full credit. The main error was using 8 or -8 as the gradient of the normal, however many candidates earned at least partial marks.

Although correct answers were seen in part (d), they were not common. Many candidates earned the method mark by the substitution of $x = 0$ and some went on to find the area of the triangle. The final mark was often lost through a failure to realise that the area was positive.

Q12.

This was the most successful question on the paper with most candidates gaining full marks or close to it. Simple sign and coefficient errors were made in the differentiation, but the vast majority of candidates demonstrated they could use the chain rule correctly. Only a few tried to find y in terms of x first, doing the work actually required for part (c). Some candidates found an equation of the tangent instead of the normal, and a few left t in their final answer. A form of the answer was not specified so any correct form was accepted, but some working with the $y = mx + c$ form sometimes found a value for c , but did not write down an equation as their final answer.

Similarly in part (c) a correct, unsimplified form was accepted for full credit, although it was noted that many did go on to make algebraic errors in attempting to simplify their equation. Most candidates did know what was required for a cartesian equation and many alternative correct versions were seen or a simple sign error was made. A few candidates thought they were required to find the tangent, and some seemed to invent their own equation, such as an ellipse, to substitute into.

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Q13.

Part (a) Almost all candidates were able to find the first and second derivatives correctly, although there was an occasional arithmetic slip; some could not cope with the fraction term, others doubled 26 incorrectly.

Part (b) Those who substituted $t = 2$ into $\frac{dy}{dt}$ did not always explain that $\frac{dy}{dt} = 0$ is the condition for a stationary point. Many used the second derivative test and concluded that the point was a maximum. Some **assumed** that a stationary point occurred when $t = 2$ and went straight to the test for maximum or minimum and only scored half of the marks.

A few tested $\frac{dy}{dt}$ on either side of $t = 2$ correctly, but those who only considered the gradient on one side of the stationary value scored no marks for the test.

(c) The concept of "rate of change" was not understood by many. Approximately equal

numbers of candidates substituted into $t = 1$ into the expression for y , $\frac{dy}{dt}$ or $\frac{d^2y}{dt^2}$ and so only about one third of the candidates were able to score any marks on this part. Those

who used $\frac{dy}{dt}$ often made careless arithmetic errors when adding three numbers.

Part (d) As in part (c), candidates did not realise which expression to use and perhaps the majority wrongly selected the second derivative. It is a general weakness that candidates do not realise that the sign of the **first** derivative indicates whether a function is increasing or decreasing at a

Q14.

This question proved to be a good source of marks for many candidates although the

expansion in part (b) was quite frequently incorrect with $\left(\frac{2}{x}\right)^2$ being written as $\frac{4}{x}$ or as $\frac{2}{x^2}$. Most candidates displayed good differentiation skills in part (c) and even more realised that the answer to part (d) required evaluation of their answer to part (c) when $x = 2$. Even though the majority of candidates knew how to find the equation of the normal it was very common to find a mark lost in the final part of the question because candidates were unable to rearrange their answer in to the form stated in the question.

Q15.

Part (a) was well answered by the majority of candidates. Many fully correct responses were seen and, if there was an error, it was usually the addition of a function of e^x or a '+ c'.

Part (b) was answered very well with many of the candidates gaining full marks, although many then went on to try and simplify their expression, which caused a loss of marks in part (c).

Some attempts at the chain rule were seen, quite often resulting in as an $\frac{x+1}{x}$ answer. Although

many correct responses were seen, part (c) was not answered as well as the two previous parts. Most of the errors came from following through from an incorrect expression in part (b). Some candidates found the gradient of the tangent and its equation but not the normal gradient. A number of candidates left their gradients in terms of x .

Q16.

Part (a)(i) was well answered by many candidates, who obtained full marks. Many often went on to obtain $e^x(x^2 + 2x - 3)$ which helped them later in the question. Part (a)(ii) was also well answered but not quite as well as part (a)(i). Many candidates obtained partial credit for reasonable attempts at the product rule but often there was a term missing.

In part (b)(i), the removal of e^x proved to be a major challenge: a large number of candidates chose to take logs (incorrectly) and, although some were lucky in the way they had written out the next step and recovered marks, many generated unmanageable expressions. In part (b)(ii), many correct responses were seen.

Q17.

This question was generally done well with most candidates demonstrating at least some knowledge and ability with implicit differentiation.

For part (a), many candidates only verified that $a = 1$ was a possible solution and thus failed to show that it was the only positive solution. Those who did form a quadratic equation as expected usually factorised it correctly, with some giving a clear argument as to why $a = 1$.

In part (b), a few candidates rearranged the given equation before attempting differentiation, and some attempted to divide through by y^2 , often making algebraic errors and leading to poor attempts at differentiation.

Most took the expression as given and were nearly always correct in differentiating the left hand side; they usually made a good attempt at either the product rule or chain rule, and often both, on the right hand side. Many fully correct solutions were seen. Errors varied

from dropping a 5 or 2, to attaching $\frac{dy}{dx}$ to the wrong term or bringing in a spurious $\frac{dy}{dx}$ having started their solution with $\frac{dy}{dx} =$. Some candidates did not evaluate their derivative at (1, 1) as requested.

In part (c), most candidates were able to use their answer to part (b) to find an equation of the tangent, although a few found an equation of the normal instead.

Q18.

(a) (i) The differentiation was generally well done, though some candidates found the fractional coefficient problematic. Those who wrote $6/3t^5$ were not

penalised in this part but writing $6 \cdot \frac{1}{3}$ generally led to errors later in the question. Some tried to avoid the fraction by considering $3V$ throughout the question, making errors, and suffered a heavy penalty.

(ii) Again, most applied the method correctly and here simplification of coefficients was necessary. A few failed to differentiate $6t$ or omitted it altogether.

(b) This part was poorly done with many not recognising that the rate of change was $\frac{dV}{dt}$ and substituted $t = 2$ into the expression for V or $\frac{d^2V}{dt^2}$. Quite a lot of candidates made arithmetic errors. A few found two values of the expression and averaged them.

- (c) (i) Again, many failed to evaluate $\frac{dV}{dt}$ in order to verify that a stationary point occurred, but those who did generally obtained a value of zero. It was essential to include a relevant statement to earn both marks.
- (ii) This required evaluation of the second derivative at $t = 1$ or an appropriate test. Candidates who tried to test the gradient on either side of 1 almost invariably failed, as the values used were too far away from the stationary point. A surprising number evaluated $10 - 24 + 6$ to be -20 thus losing the accuracy mark. Some appeared to be guessing and drew wrong conclusions about maxima or minima after evaluating the second derivative.

Q19.

- (a) Not everyone recognised that the height of the rectangle was the value of y when $x = -1$ or $x = 4$. Some who did made numerical errors. Even having found the height 4, some obtained the wrong area by taking AB as 4 or 2 (sometimes using Pythagoras' Theorem) or by finding the perimeter instead.
- (b) (i) The integral was generally correct, though sometimes incorrect simplification occurred subsequently. A few integrated x^3 to $\frac{1}{4}x^4$ and a surprising number misread the integrand as $3x^2 - x^2$. There were also candidates who confused integration with differentiation or whose process was a hybrid of the two.
- (ii) Almost everyone recognised that they should firstly evaluate the integral from -1 to 2, but most stopped there, instead of going on to subtract the value of the integral from the area of the rectangle. There were a lot of sign errors in the work with some adding instead of subtracting or putting the two parts the wrong way round. A few wrongly substituted in the original function. Those who chose to work with the differences of two integrals seldom completed it correctly.
- (c) (i) Differentiation was done well on the whole.
- (ii) Many substituted $x = 1$ into the derivative to find the gradient of the tangent and went no further. Most did not find the y coordinate of the point and so made no attempt at the equation of the tangent. A few non-linear equations were seen with a 'gradient' of $6x - 3x^2$.
- (iii) Very few candidates completed this part. Many made no attempt, and those who did tended to test a few values of x or to find the second derivative, which was of no value.

- $\frac{dy}{dx}$
 Only the strongest candidates realised that $\frac{dy}{dx} < 0$ was the condition for y to be decreasing and that, after a couple of lines of algebra, the given inequality could be obtained.
- (d) Although most made an attempt at the quadratic inequality, few obtained both parts of the solution. It was imperative that candidates wrote $x > 2$, $x < 0$ and not $0 > x > 2$

as many incorrectly stated.

It was disappointing to see how many candidates at this level could not solve the equation $x^2 - 2x = 0$, obtaining values such as -2 , $\sqrt{2}$, $1+\sqrt{2}$. Using the formula or completing the square sometimes led to $1 \pm \sqrt{1}$, which many candidates failed to simplify.

Q20.

Many candidates were able to differentiate y correctly but weaker candidates then had difficulty dealing with the negative index. The common error was to lose the negative sign in the power and write $16 - x^{-2} = 0$ as $16 - x^2 = 0$ which lead to the wrong values ± 4 . Some candidates applied the difference of two squares as the initial step in solving $16 - x^{-2} = 0$ with varying final success.

Q21.

This final question was a good source of marks for many candidates. Even the weaker candidates were able to give correct answers for parts (a) and (c) which tested standard C2 work on differentiation and integration. A significant minority of candidates started part (b)(i) by equating dy/dx to zero instead of substituting $x = 0$ to find the value of the derivative. Those who were able to answer the first two parts of part (b) correctly generally went on to find the correct coordinates of P . A significant number of candidates left their final answer as 24.3 without considering the area of the triangle OAP and carrying out the subtraction. For those candidates who attempted to find the area of the triangle some applied very lengthy methods which involved finding the length of each side of the triangle

then using the cosine rule to find angle OPA then using the formula $\frac{1}{2} ab \sin C$ or involved the integration of the equations of the two

lines. To their credit a number of such candidates did avoid premature approximation to reach the correct value for the area of the triangle which could have been obtained more

directly by using $\frac{1}{2} OA \times |y_P|$.

Q22.

The great majority of candidates knew what was expected in part (a) but some made

errors in attempting $\frac{d}{dt}\left(\frac{2}{t}\right)$; sign and coefficient errors being common and even $\ln t$ was seen.

The chain rule was usually used correctly although a few multiplied the two derivatives or had the expression 'upside down'. Errors were sometimes made in simplifying the expression, with a sign error or 2 or 8 being seen instead of $1/2$, or the t^2 term moving to the top line.

Candidates generally showed they knew how to find the equation of a straight line in part (b) using their gradient from part (a) although a few found the normal rather than the tangent. Some candidates lost a mark through not giving their final answer in the requested form, or did not notice that their a , b , c were not integers, particularly those few who had a $\ln$ term in part (a). Part (c) was usually started well but not always finished

convincingly, as the answer was given and sufficient working not shown. The question did request a verification of the given answer rather than requiring it to be shown, but few

candidates did it by the most efficient method of demonstrating $-4t \times \frac{2}{t} = -8$. There were many good answers using the elimination of t , although some candidates were too quick to write down the given answer without showing fully where it had come from.

Q23.

Part (a) Most candidates answered this part correctly. A few included the 7 or thought the derivative of the first term was $7x$ and the $-$ sign was sometimes lost.

Part (b) Many substituted $x = 1$ correctly, though it was apparent that they did not recognise this value of -10 to be the gradient of the tangent. Many correctly found y as 5 but stopped there. Again, some correct attempts at the tangent equation using $y = mx + c$ foundered and quite a large number attempted to find the equation of the normal.

Part (c) Use of the value of $\frac{dy}{dx}$ was the only acceptable method here. Evaluations of y at different points or finding the second derivative were common but earned no marks.

Q24.

Part (a)(i) Most candidates differentiated correctly. However a few made a slip or misread one of the terms.

Part (a)(ii) Although most managed to substitute 2 into their derivative some made numerical errors and some used y or the second derivative. Most who realised that they should equate their derivative to zero (or at least showed their intention though never inserting the $= 0$) then tried to factorise or use the formula (though it was clear that some did not recognise the quadratic equation as such). It was disappointing that 14 the bracket $(3x - 14)$ often produced the solution $x = 14$ instead of $14/3$, and the solution of $x = 2$ did not always appear.

Part (b)(i) The integration was also completed correctly by most candidates, although the $28x$ was occasionally wrong and some 'hybrid' processes led to terms such as $\frac{20x^3}{3}$

In part (b)(ii) almost everyone attempted to substitute 3 into their integral but their problems with the ensuing fractions often took pages to resolve, and although most ended 'magically' with the required answer there were many errors en route. A few substituted into the original expression for y instead of the integrated expression.

Part (b)(iii) This part was quite well done although again there were errors in evaluating

both $\frac{1}{2} \times 21 \times 3$ and $56 \frac{1}{4} - 31 \frac{1}{2}$. Some candidates confused length with area and merely used Pythagoras's Theorem to find the length of the hypotenuse of the triangle. Those who chose to integrate the equation of the straight line were sometimes successful but many made arithmetic errors.

Q25.

This question also proved to be a good source of marks for many candidates. The examiners expected to see some intermediate evaluation before the printed answer was quoted in part (a)(i) after the substitution of 4 for x in the given equation. Most candidates gave the correct value for the power k although the wrong values $1/2$ and $1/2$ were both seen in part (a)(ii). Candidates were generally able to differentiate the given expression to find the second derivative in part(a)(iii) but some believed that the sign of the x value rather than the sign of the second derivative determined the nature of the stationary point in (iv). In part (b) a significant number of candidates gave the equation of the tangent instead of the normal or used the gradient of the normal as -12 . Finding the integral in part (c)(i) was well answered although, somewhat surprisingly, the integration of -7 caused as many problems as the integration of the other two terms. The final part of the question proved to be beyond many average candidates who forgot to insert the constant of integration in answering the previous part. The hint 'Hence', which should have suggested the use of the answer to (c)(ii) was not always picked up. It was not uncommon to see the equation of the curve given as a linear equation, normally the tangent at P .

Q26.

Most candidates did well on this question, and even if some showed in part (a) that they apparently could not solve a quadratic equation, the implicit differentiation required in part (b) was generally done well.

Part (a) Most candidates substituted $x = 1$ correctly, and solved the resulting quadratic equation, usually by factorising, although some attempted a solution by non quadratic techniques, and usually made an error.

In Part (b) Many clear derivations were seen here, although candidates often dropped a mark through not equating their expression to zero during the derivation. The other relatively common error was in the product rule, where some candidates were not convincing in their use of signs and whether or not dy/dx was attached to a term.

Part (c) Most candidates successfully used the answer given to calculate the values of the derivatives, with some sign errors being made. Some used their own version of the derivative whereas some left the result in term of y , not realising they were to use the result of part (a).

Part (d) This part of the question defeated most candidates. Although most realised the derivative had to be set to zero, they could take it no further, and many who did write down $y - 6x = 0$ then couldn't see how to proceed. Some candidates actually "cross multiplied" to get $y - 6x = 2y - x$, usually before abandoning the attempt. Some candidates were convinced this part of the question was to do with the second derivative and did a great deal of work for no credit. Some used the given answer of $33x^2 - 5 = 0$ and solved it for x before abandoning or doing things such as substituting into the expression for the gradient.

Q27.

- (a) Most candidates knew how to differentiate, but a few made careless mistakes when differentiating $3x$ and others insisted on adding $+ C$ to their otherwise correct answer.

- (b) (i) Usually sufficient working was shown to verify that the gradient at P was equal to 5.
- (ii) Many seemed confused between the tangent and the normal, with many clearly trying to find the equation of the former rather than the latter. The major problem, once again, was rearranging the equation into one involving integer coefficients.
- (c) It was clear that some candidates were unfamiliar with the need to consider the sign of dy/dx to determine whether y was increasing or decreasing. It is a pity that $\frac{d^2y}{dx^2}$ also had value -16 when $x = 1$, yet it was surprising to see how many felt they had to consider the sign of the second derivative rather than the first.

Q28.

- (a) Once candidates realised that the lengths in centimetres of the sides of the box were $24 - 2x$, $9 - 2x$ and x , they were usually able to multiply out to give the printed answer for the volume. A large number of weaker candidates found this part of the question very difficult and made little or no effort to obtain the printed answer. (a)(i) The expression for dV/dx was usually correct.
- (ii) This part was answered too casually by many candidates who simply divided their previous answer by 12 and wrote “= 0” to comply with the printed answer. It was necessary to see that the candidate realised that it is necessary for $\frac{dV}{dx} = 0$ for stationary points to occur, before dividing their expression for dV/dx by 12.
- (iii) Most candidates scored the marks for solving the quadratic equation, although a few who chose to use the formula, rather than factorising, made slips in their arithmetic.
- (c) It was pleasing to see that most candidates were able to find a second derivative. However, this was often given as $2x - 11$, since many had previously divided dV/dx by 12 and were using this modified expression. Some candidates did not appear to understand the second derivative test for maxima and minima; they wrote things such as “ $\frac{d^2V}{dx^2} = 24x - 132 = 0$ when $x = 5.5$ and since this is positive the stationary point is a minimum”.

Q29.

Parts (a)(i) and (a)(ii) were generally well answered although it was surprising to find the derivative of x seemingly causing as many problems as the derivative of $2/x$. Although the method for finding the equation of a line was generally well known, a significant minority of candidates found the equation of the tangent rather than the equation of the normal.

Q30.

Use of the binomial expansion and use of the expansion of $(2 + x)(2 + x)^2$ were equally popular in part (a)(i) and most obtained the correct answer. There was however a minority of candidates who did not include the powers of 2 in the middle terms when applying the binomial expansion. It was disappointing to see so many candidates producing masses of algebra rather than replacing x by $(-x)$ to obtain the answer to part (a)(ii). There was some evidence of 'faking' to reach the printed answer in part (b), which quite often resulted in marks being lost as earlier correct work was changed. The vast majority realised that dy/dx was required in part (c), although some later explanations could have been more precise.



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